Predicting River Temperature and Risk from Power Plant Thermal Discharge

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Abstract

Thermal discharge from power plants can raise downstream river temperatures and stress aquatic ecosystems, especially during warm seasons. While limits on the amount of allowed heat output do exist, they are not defined dynamically to account for varying river conditions. This study uses a one-dimensional advection-diffusion-cooling model to predict how excess heat moves from a nuclear power plant through downstream transport, spreading, and atmospheric cooling. By comparing winter, spring, summer, and fall cases, the model examines whether the same permitted discharge limits produce different thermal impacts under different seasonal river conditions. The analysis includes time-dependent predictions, a steady-state decay length for plume persistence, and a population model with temperature-dependent mortality to connect temperature changes to ecological risk.

Keywords: Partial Differential Equations, Nuclear Energy, Ecological Risk, River Temperature

1. Introduction

Rivers play an important role in supporting the vitality of ecosystems by creating habitats that foster biodiversity. Many rivers play a critical role in transporting nutrients, supplying freshwater, and supporting diverse species of flora and fauna [1]. Further, river temperature is a key variable in controlling aquatic ecosystems and dynamics such as migration, determining chemical processes and understanding the physics of rivers [1]. Thus understanding river temperature is crucial in assessing the stability of ecosystems. Because of the environmental significance of rivers, there is substantial interest in documenting and modeling changes in rivers and their surrounding ecosystems, particularly in regard to temperature.

Humans also use rivers as a natural resource for many processes including irrigation, freshwater, and generating power. Power plants, especially nuclear power plants, use water to cool their systems and generate steam power for electricity-producing turbines [2]. Although vast amounts of water are used in the process of power generation, much of the water is returned to the environment in regulated ways [2, 3]. Power plants use water to absorb excess heat from the system and the water that was used in cooling becomes much hotter and is released back into the original water source.

There are many factors that can affect river temperature including natural changes, effects due to climate change and direct effects due to anthropogenic interference. Many mathematical models that study river temperature focus on natural processes from meteorological impacts and the physical geometries of rivers [4], while this paper will focus on a simplified model, assuming constant coefficients per season.

In this paper, we derive a model for predicting river temperature downstream of a power plant based on existing models. Next, we present an analytical solution to our models governing partial differential equation and use that to examine the relationship between different physical forces in the model. Then the solution will be used to predict the river temperature at a downstream location due to thermal discharge from a power plant and to assess whether the seasonal regulated output maximums will behave the same in seasonal river conditions. Lastly, the effects of these predicted seasonal river temperatures will be used in investigating the population dynamics of aquatic species.

2. Model Derivation

We propose a model to predict the river temperature downstream of a nuclear power plant that continuously discharges heat into a river. Let $T(x,t)$ be the temperature of the river at a distance x and a time t. The discharged heat moves with the flow velocity of the river u, diffuses at a rate D and experiences a heat exchange reaction between water and air at a rate per unit distance α. Power plants have a mixing zone where water is stored before it is released back into its source. We position the edge of the power plant's mixing at $x = 0$ and examine the river on the infinite half line as $x \to \infty$.

The model's governing equation is thus the advection-diffusion-reaction partial differential equation:

$$\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} = D\frac{\partial^2 T}{\partial x^2} - \alpha(T - T_e), \tag{1}$$

The reaction term for atmospheric cooling is modeled by a Newtonian cooling term, inspired by water analysis simulations from the United States Environmental Protection Program [4] and defined as

$$\frac{\partial T}{\partial t} = -\alpha(T - T_e). \tag{2}$$

Newtonian cooling defines the rate of change in river temperature as proportional to the difference between the current river temperature and its equilibrium temperature. The coefficient α is the cooling rate.

We assume that before any thermal discharge is added, the river starts at an equilibrium temperature, T_e, defined in Table 2. We then assume that the power plant, from $t > 0$, will continuously discharge heat into the river at a temperature T_{EDGE} defined in Table 1. As the heat moves through the system, the advection, diffusion and reaction terms will return the power plant's discharge back to the equilibrium as $t \to \infty$.

Thus, the model equation and boundary conditions are defined as such:

$$\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} = D\frac{\partial^2 T}{\partial x^2} - \alpha(T - T_e), \quad x > 0,\ t > 0, \tag{3a}$$

$$T(0, t) = T_{\text{EDGE}}(t), \qquad t > 0, \tag{3b}$$

$$\lim_{x\to\infty} T(x, t) = T_e, \qquad t > 0, \tag{3c}$$

$$T(x, 0) = T_e, \qquad x > 0. \tag{3d}$$

A one-dimensional model is appropriate for an approximation because we are analyzing the persistence of downstream thermal pollution, not the detailed three-dimensional mixing process.

2.1. Variables and Parameters

2.1.1 Thermal Discharge

To define the heat input from the power plant, we look at existing nuclear power plant operational permits that define a maximum temperature output. The Braidwood Nuclear Power Plant defines T_{EDGE} as the acceptable temperature for releasing the water used in the cooling process back to its original water source [3]. The Braidwood Nuclear Power Plant is located on the Kankakee River in Illinois, so we use this as a basis for our model's parameters. Table 1 outlines these temperatures monthly.

This is the direct input that our model uses to evaluate the consequences of these discharges in seasonal cases. We define the boundary condition, T_{EDGE}, as

$$T_{\text{EDGE}} = \begin{cases} 60°\text{F}, & \text{Jan., Feb., Mar., Dec.} \\ 90°\text{F}, & \text{Apr.–Nov.} \end{cases}.$$

	Jan.	Feb.	Mar.	Apr.	May	June
°F	60	60	60	90	90	90
°C	16	16	16	32	32	32
	July	Aug.	Sept.	Oct.	Nov.	Dec.
°F	90	90	90	90	90	60
°C	32	32	32	32	32	16

Table 1: Maximum allowable water temperature at the edge of the mixing zone by month published by the National Pollutant Discharge Elimination System in a permit for The Braidwood Nuclear Power Station's compliance.

2.2. Equilibrium Temperature

We assume, that before the power plant begins discharging water, the river is held at an equilibrium temperature. We also assume that the river will return to this temperature at an infinite distance. It also informs the rate a which the river temperature will return to the equilibrium from the cooling term.

Using USGS data from the past ten years at the Kankakee river [5], we present a range of possible river temperatures over the seasons. We also note the lowest recorded temperature and highest recorded temperature from the past ten years. We use this to inform extreme cases to analyze all possible river temperatures.

Table 2 shows the data average per year and season converted to Fahrenheit.

	Winter	Spring	Summer	Fall
Average (°F)	36.5	53.7	76.6	58.7
Max (°F)	43.9	69.8	83.6	74.0
Min (°F)	32.6	38.7	69.2	40.1

Table 2: Seasonal river temperature data taken from the USGS monitoring location near Wilmington, Il. This data is averaged over 11 years from 2014-2025 for each season, Winter(Dec-Feb), Spring(Mar-May), Summer(Jun-Aug), and Fall(Sep - Nov).

This is the data used in informing the T_e input in our model for each seasonal case. We also note the maximum and minimum recorded temperature for each season in the past 11 years to examine extreme cases.

2.2.1 Advection

The advection parameter is defined by a river's velocity, recorded in cubic feet per second (cfs). Because we assume a one dimensional model, we convert this to feet per

second. To do this we take our published data in cfs and divide by the cross-sectional area of the river (width×depth). This will give us a velocity in the desired units (ft/s).

Nuclear power plants are often located near large bodies of water, as required for the cooling process. Further, they are often in large, flat, stable land [2]. This is important to note, as we assume that the velocity is constant in the stretch of the river we are examining. In other words, there is no change in velocity due to changes in the terrain's gradient. **fix** Also, the size of the river constrains the amount of output from a power plant. Smaller power plants are suitable to be placed next to smaller bodies of water, while larger power plants must be placed by larger bodies of water, as noted by the U.S. Energy Information Administration [6]. For this study, we look at the output of large power plants (energy output of 1000 MW+) as their effect on the environment is much larger than that of smaller power plants.

We use published data from the past 11 years (2014-2025) from the USGS river monitoring location near Wilmington, Il, near the Braidwood power to create a range of possible river velocities [5]. The goal here is not to define an exact river velocity, but to create an acceptable range of what possible river velocities might be for power plants positioned near rivers.

To convert the river flow, given in cubic feet per second, to a value suitable for our one-dimensional model, we divide by the width and depth of the river. Records published by the National Oceanic and Atmospheric Administration indicate that the Kankakee river has a gauge between 7 ft and 15 ft [7]. Using Google Earth data, we find that the Kankakee River near the power plant is about 250 ft to 550 ft wide within 20 miles of the power plant. To get the widest range of possible advection rates, we assume for the minimum flow that the river has the lowest flow in the deepest and widest regions, and the highest flow in the shallowest and narrowest regions. We calculate the average using a width of 400 ft and a depth of 11 ft. Our results are shown in Table 3.

	Winter	Spring	Summer	Fall
Average (ft/s)	1.58	1.90	0.91	0.66
Max (ft/s)	9.04	9.98	5.65	4.71
Min (ft/s)	0.19	0.85	0.28	0.24

Table 3: Calibrated seasonal flow data in feet per second from USGS river monitoring data near Wilmington, IL from 2014-2025.

Thus, for the value of D, we use these results. This is also how we define the low flow and high flow regimes. For our seasonal analysis, Spring will have the highest river velocity, and Fall will have the lowest. We also note that maximum and minimum recorded values to further examine extreme cases.

2.2.2 Dispersion

Dispersion is the spread of a substance due to velocity. Due to hydraulic processes in rivers and turbulent flow, calculating the dispersion is a complicated task. Dispersion in a river is affected by the depth, streambed geometry, and velocity of a river. In practice, shallower and faster flow results in a higher dispersion coefficient. Additionally, in low flow regimes. Therefore we assume, based on our calculations for stream flow velocity, that the lowest dispersion regime happens in the Fall, then summer, then winter, then spring. Following hydraulic-engineering literature, D is treated as an effective longitudinal dispersion coefficient that captures bulk downstream mixing in a one-dimensional river model [8]. We take their calculated range for large rivers and convert to square feet per second to create our range of between $1\text{ft}^2/\text{s} - 16,000\,\text{ft}^2/\text{s}$ [8]. Because dispersion depends on velocity, in high flow regimes the dispersion will also be greater and in low flow regimes the dispersion will be lower.

From 3, we see that the highest average river velocity, Spring, is about three times larger than the lowest average river velocity, Fall. We assume that Spring dispersion is $7000\,\text{ft}^2/\text{s}$, and create a table from there.

	Winter	Spring	Summer	Fall
Average D (ft^2/s)	5821	7000	3353	2432
Max D (ft^2/s)	33305	36768	20816	17353
Min D (ft^2/s)	700	3132	1032	884

Table 4: Estimated seasonal dispersion coefficients scaled linearly with velocity, anchored at $D = 7000$ ft^2/s for the spring average velocity.

Assigning dispersion in this manner allows us to compare seasons effectively.

2.2.3 Atmospheric Cooling

The atmospheric cooling term represents the process at which the heated water returns to the equilibrium temperature. Modeling heat flux between water and the atmosphere takes into account many variables such as radiation, evaporation, and wind speed [9] so we start at a simplified surface flux law $H_t \simeq K(T_e - T)$ defined by The US EPA [4]. Here, H_t is the net heat flux from air to the water surface, K is the bulk atmospheric heat transfer coefficient, T_e is the equilibrium temperature and T is the actual or predicted river temperature. In a water resources research journal, Bogan et al. define the relationship between stream temperature and equilibrium temperature as Newtonian cooling [9],

$$\frac{\partial T}{\partial t} = \frac{K}{\rho \cdot c_p \cdot h}(T_e - T)$$

Thus, for our model we define the parameter

$$\alpha = \frac{K}{\rho \cdot c_p \cdot h},$$

where K is the bulk heat transfer coefficient, ρ is the density of water, c_p is the specific heat capacity of water, and h is the water depth. A broad range of K is defined by natural and forced convection in an engineering handbook [10] due to natural and extreme weather events as $K = 10 - 17000 \frac{BTU}{s \cdot\ ft^2 \cdot {}^\circ F}$ In U.S. customary units, we take $\rho \approx 62.4$ lb/ft^3, $c_p \approx 1.0$ BTU/(lb $\cdot$ °F), and $h = 11$ ft.

Substituting these values gives us the range of atmospheric cooling where

$$\boxed{\alpha \in [3.6 \times 10^{-6},\ 7.0 \times 10^{-3}]\ \mathrm{s}^{-1}.}$$

The larger α values should be interpreted as exploratory upper-bound cases. They are based on general heat-transfer estimates not river-specific calibrations, so they are used to test our model's sensitivity in cases when cooling may be stronger. Specific river calibrated α values seem to be around 5×10^{-5} s^{-1} using K values from the study of the stream temperature-equilibrium temperature relationship of Bogan et al. [9]. The specific choice of α matters less than the difference between the predicted temperature and the equilibrium temperature. The larger that difference, the heat is removed from the system.

3. Solution

To solve the partial differential we first shift the equation to

$$V = \frac{\partial V}{\partial t} + u\frac{\partial V}{\partial x} = D\frac{\partial^2 V}{\partial x^2} - \alpha V. \tag{4}$$

Employing this shift $V(x,t)$ now represents the excess temperature in the system, or the temperature above equilibrium. After this we can then use the Laplace transform to turn the problem into an ODE in Laplace space. This problem is ripe for the Laplace transform given the initial condition at $t = 0$, and because we are dealing with a linear partial differential equation.

The Laplace Transform on $V(x,t)$ is defined as

$$\bar{V}(x,s) = \mathcal{L}\{V(x,t)\} = \int_0^\infty e^{-st} V(x,t)\, dt.$$

Applying this transform to each term in the partial differential equation and on the initial and boundary conditions, see Appendix, we obtain an ODE in $\bar{V}(x,t)$

$$D\bar{V}_{xx} - u\bar{V}_x - (s+\alpha)\bar{V} = 0,$$

which we can solve with the general exponential form solution. Then, using the initial conditions, we obtain our solution in Transformed space,

$$\bar{V}(x,s) = \frac{V_{\text{edge}}}{s} \exp\left(\frac{u - \sqrt{u^2 + 4D(s+\alpha)}}{2D} x\right).$$

Transforming back, gives us the analytical solution in terms of $V(x,t)$

$$\boxed{V(x,t) = \frac{V_{\text{edge}}}{2}\left[e^{\frac{u-\beta}{2D}x}\operatorname{erfc}\left(\frac{x-\beta t}{2\sqrt{Dt}}\right) + e^{\frac{u+\beta}{2D}x}\operatorname{erfc}\left(\frac{x+\beta t}{2\sqrt{Dt}}\right)\right]} \tag{5}$$

$$\beta = \sqrt{u^2 + 4\alpha D}. \tag{6}$$

And shifting back to the original temperature problem

$$\boxed{T(x,t) = T_e + \frac{T_{\text{edge}} - T_e}{2}\left[e^{\frac{u-\beta}{2D}x}\operatorname{erfc}\left(\frac{x-\beta t}{2\sqrt{Dt}}\right) + e^{\frac{u+\beta}{2D}x}\operatorname{erfc}\left(\frac{x+\beta t}{2\sqrt{Dt}}\right)\right],} \tag{7}$$

This expression forms the basis for the downstream temperature profiles and parameter studies presented in the following sections.

4. Results

4.1. Interaction of Transport and Cooling Mechanisms

The analytical solution in Eq. 7 makes it possible to compare how the physical parameters of the model influence the downstream thermal plume. We use the solution to evaluate the excess temperature,

$$T(x,t) - T_e,$$

at a fixed downstream location and time. This allows the model to ask a more direct physical question: under what combinations of river velocity, dispersion, and atmospheric cooling does heated water still remain at a downstream point?

For this comparison, the excess temperature is evaluated at $x = 13000$ ft and $t = 2.0$ hr. The location $x = 13000$ft represents a fixed downstream point at which thermal exposure may matter for aquatic organisms, while the fixed time allows the effects of the advection parameters to be compared on the same scale. The heat maps therefore show the sensitivity of the predicted temperature at one downstream location to changes in the model parameters.

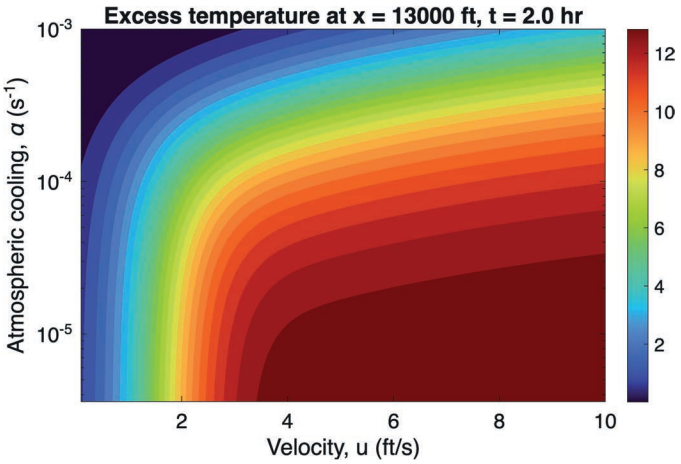


Figure 1: Heat map of excess temperature $T(x,t) - T_e$ at $x = 13000$ ft and $t = 2.0$ hr as a function of river velocity u and atmospheric cooling rate α. Warmer colors indicate larger excess temperatures above the equilibrium river temperature.

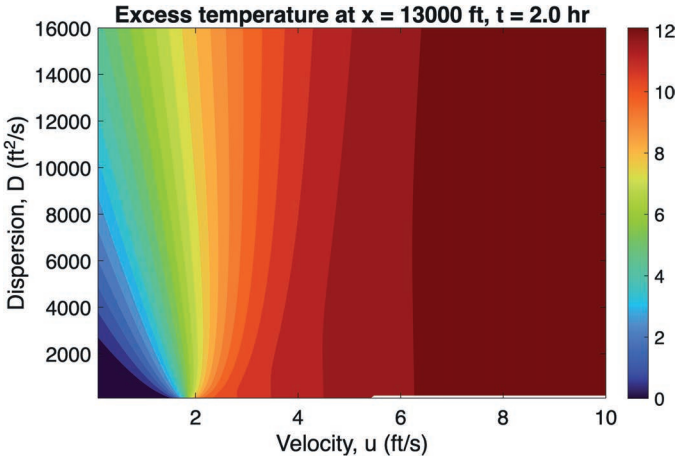


Figure 2: Heat map of excess temperature $T(x,t) - T_e$ at $x = 13000$ ft and $t = 2.0$ hr as a function of river velocity u and longitudinal dispersion coefficient D.

Figure 1 shows the interaction between river velocity and atmospheric cooling. Larger u values increase the excess temperature at the downstream point because heated water is transported to $x = 13000$ ft more quickly. In contrast, larger α values decrease the excess temperature because the plume loses heat to the atmosphere before reaching that location. The highest downstream temperatures occur when the river velocity is high and the atmospheric cooling rate is low. In that case, advection moves heat downstream faster than atmospheric exchange can remove it. The lowest downstream temperature occurs when the cooling rate is large and velocity is small, because heat is dissipated before much of the plume reaches the fixed point.

Figure 2 shows how velocity and longitudinal dispersion control downstream heat transport. When velocity is low, heated water does not move quickly enough to reach the downstream point by advection alone. In this region of the parameter space, increasing D raises the excess temperature at $x = 13000$ ft because dispersion spreads heat farther downstream. When velocity is high, the river flow carries heat to the downstream point, so excess temperature remains high even when dispersion is moderate. The largest amount of excess temperatures occur when both velocity and dispersion are high, because advection transports the plume downstream while dispersion widens the region affected by the thermal discharge.

Together, Figures 1 and 2 show that impact thermal discharge cannot be determined from just the edge of mixing zone temperature. The same value of T_{EDGE} can lead to different downstream temperatures depending on the river velocity, dispersion, and atmospheric cooling rate. This motivates the seasonal simulations that follow. Because winter, spring, summer, and fall have different ambient river temperatures and flow conditions, a fixed regulatory discharge limit may not correspond to a standard downstream temperature.

4.2. Are the permitted thermal discharge limits effective in controlling downstream temperature?

Using Eq. 7, we examine the long-time behavior of the thermal plume. This steady-state limit is useful because it describes how river temperature behaves in each season in the long run. As $t \to \infty$,

$$\frac{x - \beta t}{2\sqrt{Dt}} \to -\infty,$$

so

$$\operatorname{erfc}\left(\frac{x - \beta t}{2\sqrt{Dt}}\right) \to 2.$$

Similarly,

$$\frac{x + \beta t}{2\sqrt{Dt}} \to +\infty,$$

so

$$\operatorname{erfc}\left(\frac{x + \beta t}{2\sqrt{Dt}}\right) \to 0.$$

Therefore, the steady-state temperature profile is

$$\boxed{T_{ss}(x) = T_e + (T_{edge} - T_e)e^{\frac{u-\beta}{2D}x}} \tag{8}$$

where

$$\beta = \sqrt{u^2 + 4\alpha D}.$$

Because $\beta > u$, the exponent is negative, meaning that the excess temperature decays downstream. To examine the decay of the thermal plume, we define

$$\kappa = \frac{\beta - u}{2D} = \frac{\sqrt{u^2 + 4\alpha D} - u}{2D}.$$

Then the steady-state solution can be written as

$$T_{ss}(x) = T_e + (T_{edge} - T_e)e^{-\kappa x}. \tag{9}$$

Because we impose the condition that the river will return to the equilibrium as $x \to \infty$, the steady state will asymptotically approach the equilibrium for each season. So, a more interesting quantity is the length it takes to approach the equilibrium.

Equation 9 shows that the steady-state excess temperature, decays exponentially with distance downstream. This allows the thermal plume to be summarized by a characteristic decay length,

$$\lambda = \frac{1}{\kappa} = \frac{2D}{\sqrt{u^2 + 4\alpha D} - u}.$$

Using this notation, the steady-state solution becomes

$$T_{ss}(x) = T_e + (T_{edge} - T_e)e^{-x/\lambda}.$$

The characteristic decay length, $x = \lambda$, is the distance at which the excess temperature has decreased to $e^{-1} \approx 36.8\%$ of its value at the edge of the mixing zone. Thus, larger values of λ indicate that heat persists farther downstream, while smaller values of λ indicate faster thermal recovery.

To compare how far excess heat persists downstream across seasons, we normalize the steady-state temperature profiles by the initial excess temperature at the edge of the mixing zone.

So we define the normalized excess temperature

$$\Theta(x) = \frac{T_{ss}(x) - T_e}{T_{\text{EDGE}} - T_e}.$$

Substituting the steady-state solution gives

$$\Theta(x) = e^{\left(-\frac{x}{\lambda}\right)}.$$

This normalization makes comparisons between seasons easier. Figure 3 shows the downstream persistence of the thermal plume itself. Larger values of λ correspond to slower thermal decay and therefore excess heat travels farther downstream.

The dashed vertical markers in Figure 3 show this characteristic distance for each season, while the horizontal reference line marks when $x = \lambda$ or equivalently $\Theta = e^{-1}$.

Figure 3 shows that the seasons differ how excess heat is transported and dissipated downstream. Because the normalized profiles remove the variations between seasonal temperatures, the remaining differences are then only controlled by the decay length λ, which depends on velocity, dispersion, and atmospheric cooling. The decay length provides a compact way to compare the impact of thermal discharge across seasons and to identify in which seasons a river experiences the largest impact from the power plant.

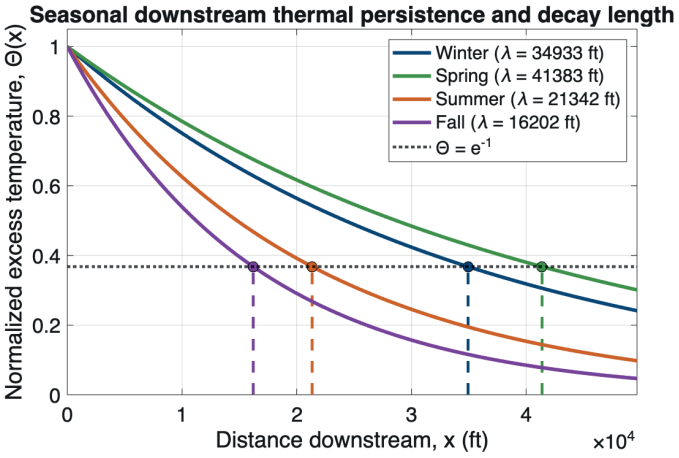


Figure 3: Normalized steady-state excess temperature profiles, $\Theta(x)$, for winter, spring, summer, and fall, computed using seasonal values from Tables 2–4. The dashed horizontal line marks $\Theta = e^{-1}$, and the vertical dashed lines show the seasonal decay lengths: $\lambda = 34{,}933$ ft winter, 41,383 ft spring, 21,342 ft summer, and 16,202 ft fall.

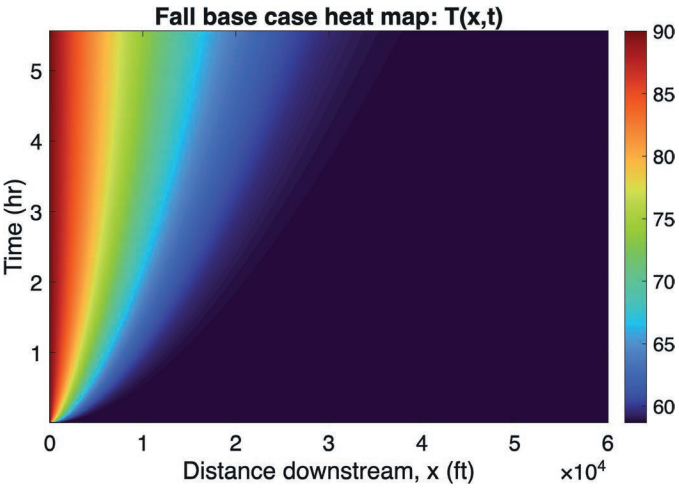


Figure 4: Base-case heat map for the fall thermal plume showing $T(x,t)$ in degrees Fahrenheit over $0 \le x \le 6.0 \times 10^4$ ft and approximately $0 \le t \le 5.5$ hours. The fall parameters are chose from the parameter tables.

4.3. How can we offset seasonal plume persistence?

We have shown how seasonal river parameters produce different downstream temperatures. Because equilibrium temperature and advection, dispersion, and cooling rates are governed by natural forces, we cannot change those to control river temperature. Thus we can change the one anthropogenic value, T_{EDGE}. Figures 4 and 5 further illustrate how temperature is not consistent at a certain distance at each season. We can use this to set a standard down stream temperature across seasons because fall has the shortest decay length.

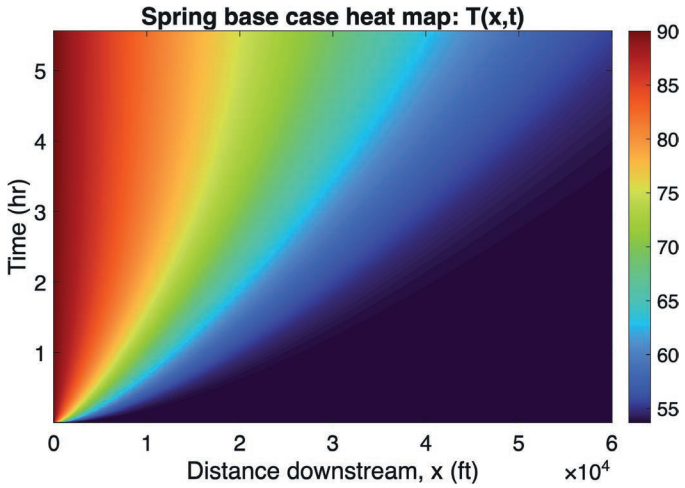


Figure 5: Base-case heat map for the spring thermal plume showing $T(x,t)$ in degrees Fahrenheit over $0 \le x \le 6.0 \times 10^4$ ft and approximately $0 \le t \le 5.5$ hours. The spring case uses parameters outlined in the parameter tables.

The base case heat map shows how temperature changes in both distance and time. This gives the overall picture of how thermal discharge from a power plant behaves in a river. If we use fall's steady-state profile as the standard baseline, we can invent an inverse problem to solve for T_{EDGE} at specific locations to match the temperature of the fall regime. Thus, we impose the matching condition $T_{ss,s}(x_0) = T_f(x_0)$, where the subscript s and f denote spring and fall respectively. Using the spring steady-state solution we solve for the required spring T_{EDGE} and obtain

$$T^{\text{req}}_{\text{EDGE},s}(x_0) = T_{e,s} + (T_f(x_0) - T_{e,s})\, e^{\left(\frac{x_0}{\lambda_s}\right)}.$$

This expression defines an inverse matching function. For each possible downstream matching location x_0, it gives the spring boundary temperature needed for the spring steady-state temperature to equal the fall steady-state temperature at that specific location. In Figure 6, we use $x^* = 13{,}000$ ft, which gives

$T^{\text{req}}_{\text{EDGE},s}(13{,}000) = 79.75°\text{F}$.

Figure 6 shows the inverse calculation needed to make spring behave like fall at a specified downstream location. The original spring profile remains warmer than the fall profile downstream because spring has a larger decay length, meaning that excess heat is transported farther before decaying toward equilibrium.

The adjusted spring curve demonstrates that lowering the spring boundary temperature can create agreement with fall at one selected downstream location. However, because spring and fall have different velocities, dispersion coefficients, equilibrium temperatures, and decay lengths, one constant value of $T_{\text{EDGE},s}$ cannot make the entire spring profile identical to the fall profile. So, we can use the function $T_{\text{EDGE}}(x)$ and select a certain distance that cannot be raised above a specified temperature. This provides a new method for calculating thermal discharge temperature maximums per season.

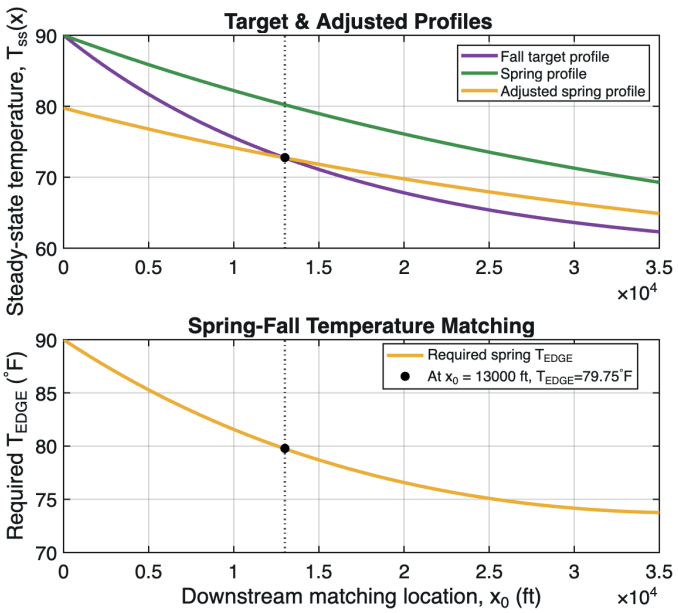


Figure 6: Plot of the inverse steady-state comparison between fall and spring thermal plume behavior. The upper panel shows the fall target profile, the original spring profile with $T_{\text{EDGE}} = 90°\text{F}$, and an adjusted spring profile chosen to match the fall temperature at $x_0 = 13{,}000$ ft. The lower panel shows the required spring T_{EDGE} temperature, needed to match the fall steady-state temperature at each downstream location. At $x_0 = 13{,}000$ ft, spring requires $T_{\text{EDGE}} = 79.75°\text{F}$ to match the fall profile.

4.4. Consumer–Resource Model

As discusses earlier, river temperature plays a vital role in supporting aquatic ecosystems. To see how the temperature model affects ecological dynamics, we study a consumer–resource ordinary differential equation model [11] in which phytoplankton act as the resource and mussels act as the consumer. Under natural settings, populations often find their own equilibrium, however, we asses how human-impact can disrupt these natural states. These two species are chosen as they remain stationary over time in a river, meaning they cannot relocate when river temperature changes. We use a standard consumer–resource model with logistic resource growth and a Type I functional. We assume that the phytoplankton observes logistic growth so they do not overtake the whole river ecosystem. Further, we assume a Type I functional response, so that rate of the mussels' consumption increases linearly with the population of the resource, phytoplankton. So, when there are more phytoplankton, the mussel population will

increase. We choose these dynamics to examine the effect of river temperature because of the relationship between phytoplankton growth and mussel mortality have an opposite response to excess heat. The model is then defined as

$$\frac{dP}{dt} = r_P(T)P\left(1 - \frac{P}{K_P}\right) - gPM, \tag{10}$$

$$\frac{dM}{dt} = egPM - m_M(T)M. \tag{11}$$

Here, $P(t)$ represents the resource population, phytoplankton, and $M(t)$ represents the consumer population, mussels. The parameter $r_P(T)$ is the temperature growth rate of phytoplankton, K is the phytoplankton carrying capacity, g is the grazing rate, e is the conversion efficiency from consumed resources into consumer growth, and $m_M(T)$ is the temperature dependent mussel mortality rate. The grazing term gRC represents a Type I functional response, meaning that consumption is proportional to both resource abundance and consumer abundance. To incorporate the effect of thermal discharge, we allow the phytoplankton growth rate and mussel mortality rate to depend on the temperature predicted by the thermal plume model.

The temperature-dependent phytoplankton growth rate is modeled as

$$r_P(T) = r_{P,0} + a_P \max(0, T - T_P^*),$$

where $r_{P,0}$ is the baseline phytoplankton growth rate, a_P controls how strongly phytoplankton growth increases with temperature, and T_P^* is the temperature threshold above which phytoplankton growth begins to increase. Similarly, the temperature-dependent mussel mortality rate is modeled as

$$m_M(T) = m_{M,0} + a_M \max(0, T - T_M^*),$$

where $m_{M,0}$ is the baseline mortality rate for mussels, a_M controls how strongly mussel mortality increases with temperature, and T_M^* is the temperature threshold above which mussels experience additional thermal stress. This model assumes mortality and growth rates to remain at a baseline level below the temperature threshold and then increase approximately linearly with excess temperature as a simple way to represent temperature dependence [12].

Because the system is temperature dependent, the equilibrium will change as temperature changes. So we define an instantaneous equilibrium for the steady-state analysis. Then $r_P(T)$ and $m_M(T)$ are constants for that temperature. The model has three biologically relevant equilibrium, see Appendix for calculation:

$$E_0 = (0, 0),$$
$$E_P = (K_P, 0),$$
$$E_C = \left(\frac{m_M(T)}{eg}, \frac{r_P(T)}{g}\left[1 - \frac{m_M(T)}{egK_P}\right]\right).$$

The extinction equilibrium $E_0 = (0, 0)$ is unstable. If a small amount of phytoplankton are introduced in the absence of mussels the the phytoplankton population grows. In the absence of phytoplankton, if mussels are introduced, they will have no food source and then die off. The phytoplankton-only equilibrium $E_P = (K_P, 0)$ is stable when

$$m_M(T) > egK_P.$$

In this case, mussel mortality is too large for mussels to persist. There is no mussel-only equilibrium because in the absence of phytoplankton, mussels will die off.

The coexistence equilibrium E_C is biologically meaningful only when both populations are positive, which requires

$$m_M(T) < egK_P.$$

When this condition holds, the coexistence equilibrium is stable. Therefore, the main mussel persistence threshold is

$$\boxed{m_M(T) = egK_P.}$$

This threshold represents a bifurcation between the phytoplankton-only equilibrium and the coexistence equilibrium. When $m_M(T) < egK_P$, mussels and phytoplankton coexist. When $m_M(T) > egK_P$, the coexistence equilibrium is lost and the model predicts a transition toward a phytoplankton-only state.

Thus, the thermal plume model affects the ecological dynamics by changing $r_P(T)$ and $m_M(T)$. The phytoplankton growth rate $r_P(T)$ influences the size of the coexistence equilibrium, while the mussel mortality rate $m_M(T)$ determines whether mussels can persist. When the predicted temperature is high enough that $m_M(T) > egK_P$, the model predicts a shift from stable coexistence to a decline in mussel population.

Even though we model each season over a 90 day period, for the phase plane portrait we examine the long term behavior and conclude that thermal discharge from a power plant does fundamentally change the population dynamics of the species. Figures 7 and 8 show how the thermal plume model can be used as an input into a simple ecological response model. The purpose of this extension is not to make a calibrated prediction of mussel or phytoplankton population, but to demonstrate how seasonal differences and anthropogenic impact in river temperature can alter the qualitative behavior of an ecosystem. For that reason, the population variables are normalized and the ecological parameters are chosen to produce interpretable results.

The main finding is that regimes with less heated added from the power plant allow the mussel and phytoplankton populations to approach a balanced coexistence state, while the hottest temperatures cause the mussels to die. In the Figure 8, this dynamic can be seen between seasons. Under cooler conditions, mussels survive and phytoplankton settle toward an steady level. However, under

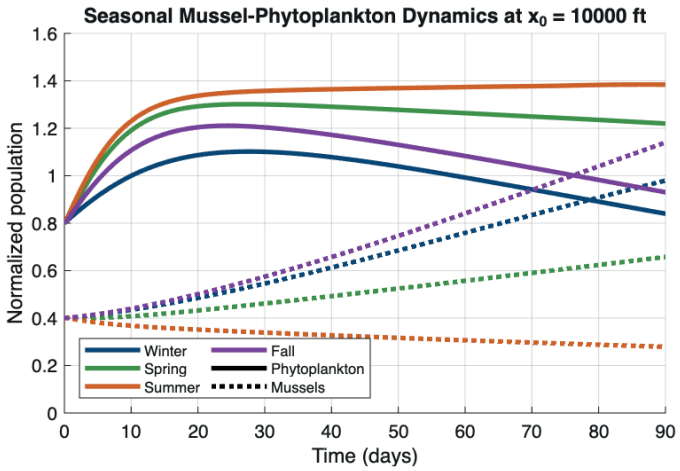


Figure 7: Seasonal phytoplankton–mussel dynamics at $x_0 = 10000$ ft over 90 days. Solid curves show phytoplankton $P(t)$ and dotted curves show mussels $M(t)$; color indicates season. Parameters: $r_{P,0} = 0.12$ day^{-1}, $a_P = 0.006$ day^{-1} °F^{-1}, $T_P^* = 68$°F, $K_P = 1.5$, $m_{M,0} = 0.008$ day^{-1}, $a_M = 0.004$ day^{-1} °F^{-1}, $T_M^* = 80$°F, $g = 0.06$ day^{-1}, $e = 0.30$, $P(0) = 0.8$, and $M(0) = 0.4$. Seasonal differences arise from the PDE-predicted temperature input $T(x_0, t)$.

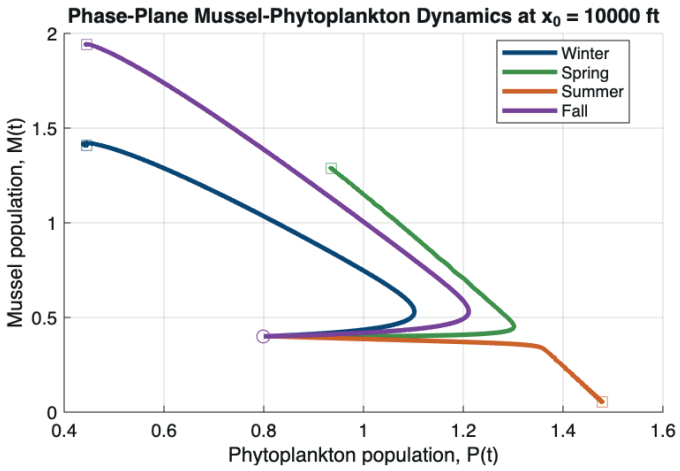


Figure 8: Phase-plane trajectories for the seasonal phytoplankton–mussel model. Each curve shows the ecological state $(P(t), M(t))$ over the same long term simulations shown in Figure 7. Circles mark the initial state $(0.8, 0.4)$ and squares mark the final state. The parameters are the same as in Figure 7.

warmer conditions, the mussel population decreases as the temperature-dependent mortality rate becomes stronger. Trajectories that move toward an inner region on the phase-plane portrait represent coexistence between phytoplankton and mussels, while trajectories that move downward represent a loss of mussels from the system.

This behavior is consistent with the steady-state condition derived above. For a fixed temperature, mussel persistence requires

$$m_M(T) < egK_P.$$

The hottest season produces mussel decline because the thermal plume raises $m_M(T)$ above the persistence threshold, while cooler seasons remain below that threshold and allow coexistence. Because summer is the hottest season, at this specific location $x_0 = 100,000$ ft

The assumptions produce a model in which river temperature changes the balance between resource growth, consumer grazing, and consumer mortality. These plots emphasize that river temperature is an important ecological variable that can have strong impacts on aquatic ecosystems. Seasonal equilibrium temperature, flow velocity, dispersion, and atmospheric cooling determine how warm the river remains downstream of the discharge point. Those predicted temperatures then affect the biological rates in the ecological model. As a result, the same thermal discharge can have different ecological consequences depending on season. This suggests that thermal discharge limits should be interpreted beyond just downstream temperature profiles and factor in how those temperatures interact with ecosystems.

5. Conclusion

In this study we developed a simplified model for predicting river temperature downstream of a nuclear power plant using a one-dimensional advection–diffusion–reaction equation. This approach connects to existing stream temperature models that use heat transport and heat-exchange terms to describe river temperature dynamics [13, 4, 9]. Unlike more detailed heat models, this project uses an analytical framework to isolate how advection, dispersion, atmospheric cooling, and seasonal conditions shape downstream thermal behavior.

The results show that seasonal conditions produce different downstream temperature responses, even when the same discharge temperature is used. This matters because the permit conditions considered in this study specify only two seasonal discharge limits: 60°F from December through March and θ °F from April through November [3]. These values provide useful boundary conditions, but they do not fully account for seasonal changes in background river temperature, velocity, dispersion, or cooling capacity.

To connect the physical model to ecosystem response, the predicted temperature was used as an input into a temperature-dependent consumer–resource model with phytoplankton as the resource and mussels as the consumer. This supports previous work showing that river temperature is closely tied to aquatic ecosystem processes and biological response [14].

Future work should use time-dependent data for river flow, background temperature, atmospheric forcing, and discharge conditions. The same framework could also test alternative discharge strategies, such as adjusting boundary temperatures seasonally or dynamically to keep downstream temperatures below ecologically meaningful thresholds. Overall, this study shows that thermal discharge is more complex than a physical heat-transport problem and can be thought as a seasonal ecological problem.

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Bailey Parent

What are we optimizing for?



That question follows me everywhere.
From my studies in math, I am no stranger to optimizing. I have become fluent in the language of objective functions, inputs, and constraints. Even my decision to study math came from a kind of optimization: if I could choose one major, which would leave the most doors open afterward? Math seemed to maximize opportunity. But my earliest optimization problems were not in math. As an Eagle Scout, I learned you don't only optimize a pack to make it light for you; you optimize it so the group can ultimately finish the trek. I learned that the best solution is not always the one that benefits a singular person, rather it is the one that makes the system more survivable for everyone in it. That lesson shaped the way I entered math, even though it was not an obvious path for me at first. I came to college intending to be a Classics major, as I was more drawn to interpretation than calculation and more comfortable in subjective spaces than technical ones. Over time I realized that what I loved about the humanities was also present in math. That idea changed how I understood technical work. Numbers and equations do not stand apart from the human questions that interested me; they were just another way to interpret them. By the time I moved more seriously into applied mathematics, I saw math as the language to begin solving these human problems.
That is the mindset I bring to this project. A model of heat output can explain how energy moves from a power plant into a river, but it does not automatically account for nearby communities or environmental risk. Mathematics gives me tools to study complex systems, but the non-math parts of me determine what I choose to study, and why.
So when I ask, What are we optimizing for? I am no longer asking only a mathematical question. I am asking an ethical one.